

Paper for USRA 2025 - Studing the Wiener Index for spatially embedded graphs

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Abstract

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1 Introduction

The Wiener index is a topological index that is defined as the sum of distances between all pairs of vertices in a graph. It is widely used in chemistry to predict the properties of molecules based on their structure. In this paper, we will study the Wiener index for spatially embedded graphs, which are graphs that can be drawn in a Euclidean space.

Mathematically, the *Wiener index* of a weighted graph $G = (V, E)$ is the sum of the distances between all pairs of vertices in G :

$$W(G) = \sum_{u,v \in V} \delta_G(u, v)$$

Where $\delta_G(u, v)$ is the weight of the shortest (minimum-weight) path between vertices u and v in the graph G . Formally, the Wiener index was introduced by the chemist Harry Wiener in 1947 [2]. It hand its variations have several applications in chemistry, biology, and network analysis.

2 Preliminaries

Let P be a set of n points in the Euclidean plane $\mathbb{R}^2$. The *Euclidean distance* between two points $p_i, p_j \in P$ is defined as:

$$|p_i p_j| = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Let $G = (P, E)$ be the complete graph on the set of points P , where the edge set E consists of all pairs of points in P . The weight of an edge $(p_i, p_j) \in E$ is defined as the Euclidean distance between the points p_i and p_j and is denoted as $w(p_i, p_j) = |p_i p_j|$.

Let T be a spanning tree of the graph G (That is, it spans all the points in P and is connected). For each pair of points $(p, q) \in P$, let $\delta_T(p, q)$ be the weight of the (unique) shortest path between p and q in the tree T .

Then, we define the *Wiener index* of the tree T as:

$$W(T) = \sum_{p,q \in P} \delta_T(p, q)$$

Finally, we define the total weight of the paths in T from p to every other point $q \in P$ as:

$$\delta_p(T) = \sum_{q \in P} \delta_T(p, q)$$

Add citation about the various applications here

Add more details about the Wiener index, its properties, a brief overview about the work done to find the optimal way to compute it for different graphs, and the challenges in computing it for spatially embedded graphs.

Add a section about our contribution here.

3 Computing the Wiener index for spatially embedded graphs

Abuaffash et al. [1] showed the following theorems:

Theorem 3.1 (Abuaffash et al. 2023). *The spanning tree of P that minimizes the Wiener index is planar.*

Theorem 3.2 (Abuaffash et al. 2023). *There exists a $O(n^4)$ time algorithm that constructs a spanning tree of P that minimizes the Wiener index.*

Theorem 3.3 (Abuaffash et al. 2023). *Given a cost W and a budget B , computing a spanning tree of P whose Wiener index is at most W and whose total weight of the paths from each point to every other point is at most B is weakly NP-hard.*

Theorem 3.4 (Abuaffash et al. 2023). *The Hamiltonian path of P that minimizes the Wiener index is not necessarily planar and computing it is NP-hard.*

We tackle the problem of computing the Wiener index for points in general position as well as give an approximation algorithm for the problem of computing the hamiltonian path of P that minimizes the Wiener index.

4 Hardness reduction

In *Geometric Spanning Trees Minimizing the Wiener Index* (A. Karim Abu-Affash), one of the listed open questions is:

Another challenging open question is to determine the complexity of computing a spanning tree of minimum Wiener index for a set P of n points in general position: Is this problem NP-hard? In this paper, we considered a variant of this problem when the total weight of the tree is bounded, and we proved that this variant is NP-hard.

We are interested in the unconstrained Euclidean version (no bound on total weight). The conjecture is that it is NP-hard.

4.1 Identity for Wiener Index of a Tree

Let $T = (P, E)$ be a tree on $n = |P|$ vertices with edge lengths $\ell : E \rightarrow \mathbb{R}_{>0}$. For an edge $e \in E$, let s_e be the size of one of the two components obtained by removing e (the product $s_e(n - s_e)$ is symmetric, so the choice does not matter).

Theorem 4.1 (Edge Decomposition Formula).

$$\text{WI}(T) = \sum_{\{u,v\} \subseteq P} \text{dist}_T(u, v) = \sum_{e \in E} \ell(e) s_e (n - s_e).$$

Proof. For each edge $e \in E$ and unordered pair $\{u, v\} \subseteq P$, define

$$I_e(u, v) = \begin{cases} 1, & \text{if } e \text{ lies on the unique } u-v \text{ path in } T, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\text{dist}_T(u, v) = \sum_{e \in E} \ell(e) I_e(u, v).$$

Summing over all unordered pairs and exchanging summations,

$$\sum_{\{u,v\} \subseteq P} \text{dist}_T(u, v) = \sum_{e \in E} \ell(e) \sum_{\{u,v\} \subseteq P} I_e(u, v).$$

When e is removed, the tree splits into two components of sizes s_e and $n - s_e$. A pair $\{u, v\}$ uses e iff u and v lie in different components. There are exactly $s_e(n - s_e)$ such pairs. Thus

$$\sum_{\{u,v\} \subseteq P} \text{dist}_T(u, v) = \sum_{e \in E} \ell(e) s_e (n - s_e).$$

□

Corollary 4.2 (Weighted version). *If each vertex $v \in P$ has weight $w(v) \geq 0$, then the weighted Wiener index*

$$\text{WWI}(T) = \sum_{\{u,v\} \subseteq P} w(u)w(v) \text{dist}_T(u, v)$$

satisfies

$$\text{WWI}(T) = \sum_{e \in E} \ell(e) W_e (W_{\text{tot}} - W_e),$$

where W_e is the sum of vertex weights in one component after removing e and $W_{\text{tot}} = \sum_{v \in P} w(v)$.

4.2 Reduction Blueprint for NP-Hardness

We sketch an NP-hardness reduction using the edge-decomposition identity. The goal is to reduce from 3-PARTITION.

4.2.1 Key Insight

The Wiener index of a tree decomposes as

$$\text{WI}(T) = \sum_{e \in E} \ell(e) s_e (n - s_e).$$

Thus:

- Long edges are expensive only if they cut many pairs.
- By embedding points with widely separated scales, we can force a spanning tree to have a prescribed *backbone* with exactly one long edge per large gap.

4.3 Reduction Sketch (from 3-PARTITION)

- **Backbone:** Place $m + 1$ hub points $H_0, \dots, H_m$ along the x -axis, with gaps $L_1 \ll L_2 \ll \dots \ll L_m$ where $L_{k+1} \geq n^4 L_k$ (scale separation).
- **Item Gadgets:** For each integer a_i , create a cluster C_i of a_i points in a tiny ball of radius ε . Clusters are positioned so they can connect cheaply to one of two adjacent hubs, but crossing a large gap is very expensive.
- **Unique Edge per Gap:** By scale separation, any optimal tree must use exactly one edge per gap (otherwise the extra long edge would dominate the objective).
- **Lexicographic Minimization:** The Wiener index reduces to

$$\text{WI}(T) \approx \sum_{k=1}^m L_k S_k (n - S_k),$$

where S_k is the number of points to the left of gap k . Because $L_{k+1} \gg L_k$, minimizing WI is lexicographic in the sequence $(S_1, \dots, S_m)$.

- **Bin Packing Interpretation:** The reduction enforces that each gap corresponds to one *bin* which must hold exactly three item clusters. To minimize the Wiener index, each bin must sum to the same target B ; otherwise S_k deviates, incurring a penalty at scale L_k . This encodes 3-PARTITION.

4.4 Weaker Variant

A simpler reduction from PARTITION is possible with only one large gap: minimizing

$$L \cdot s(n - s)$$

is equivalent to balancing two sides of size $n/2$, directly encoding a partition of item weights.

5 References

- A. Karim Abu-Affash, “Geometric Spanning Trees Minimizing the Wiener Index.”
- I. Gutman, “The Wiener Index of Trees and Its Generalizations,” surveys (various).
- A. A. Dobrynin, R. Entringer, I. Gutman, “Wiener index of trees: Theory and applications,” *Acta Applicandae Mathematicae*, 66 (2001), 211–249.
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6 Conclusion

In this paper, we studied whatever that we needed to study :).

References

- [1] A. Karim Abu-Affash, Paz Carmi, Ori Luwisch, and Joseph S. B. Mitchell. Geometric spanning trees minimizing the wiener index, 2023.
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